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MILITARY SPECIFICATION

MEATLOAF WITH MUSHROOM GRAVY, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers meatloaf with mushroom gravy, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8940

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MIL-L-1497	- Labeling of Metal Cans for Subsistence Items
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Personal Equipment and Equipage; General Specification For
MIL-C-44340	- Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-900	- Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies may be obtained from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Cereal Panel Odor and Flavor Evaluation System

(Copies may be obtained from the Northern Regional Research Center (ARS-USDA), 1815 North University, Peoria, IL 61604.)

Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies may be obtained from the Director, Livestock and Seed Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2092, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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Meat and Poultry Inspection Regulations

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

Regulations Governing the Inspection of Eggs and Egg Products

(Copies may be obtained from the Director, Poultry Division, Agricultural Marketing Service, U.S. Department of Agriculture, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Condition of Food Containers

(Copies may be obtained from the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, U.S. Department of Agriculture, Room 0610, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Grades of Nonfat Dry Milk

(Copies may be obtained from the Chief, Dairy Standardization Section, Dairy Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2750, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. Standards for Grades of Tomato Paste
U.S. Standards for Grades of Canned Mushrooms

(Copies may be obtained from the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Copies should be obtained from the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Copies should be obtained from the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Copies should be obtained from the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Copies should be obtained from the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any combination of carcasses, quarters, and any of the following recognizable primal and subprimal cuts or portions thereof: square-cut chucks, shoulder clods, chuck rolls, ribs, ribeyes, trimmed full loins, trimmed short loins, strip loins, sirloins, top sirloin butts, rounds, top rounds, peeled knuckles, outside rounds, and bottom rounds (heel out). Recognizable cuts are those which, when compared to Institutional Meat Purchase Specifications (IMPS) cuts, have no more than a minor amount of lean, fat, or bone removed or included from an adjacent cut. A portion is a certified recognizable cut from which objectionable tissue, such as tendons or heavy connective tissue, has been removed. Tenderloins, rib fingers (intercostal meat), and any part of a rough cut may be excluded at the option of the contractor. The beef shall be in the fresh chilled state and shall be in excellent condition; i.e., exposed lean and fat surfaces shall be of a color and bloom normally associated with the class and cut of meat, and typical of meat that has been properly stored and handled. Cut surfaces and naturally exposed lean surfaces shall show no more than slight darkening or discoloration due to dehydration, aging, or microbial activity. The fat shall show not more than a very slight discoloration due to oxidation or microbial activity. No odors foreign to fresh meat shall be present. Changes in color and odor characteristically associated with vacuum-packaged meat in excellent condition shall be acceptable. Also, meat shall show no evidence of freezing, defrosting, or mishandling.

3.2.1.1 Boning and trimming. The beef shall be boned and trimmed to remove objectionable material such as bone, cartilage, heavy connective tissue, etc. The boneless trimmed beef shall meet the limitations specified in tables I and II.

3.2.1.2 Handling and storage. Handling and storage of the boned and trimmed beef prior to processing into the finished product shall be in accordance with the following requirements:

a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28° to 50°F (inclusive).

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b. Holding in the fresh chilled state for not more than 4 days after initial certification is permitted provided that the beef is maintained in the temperature range of 28° to 40°F (inclusive).

c. Holding in the frozen state for not more than 120 days after initial certification is permitted provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence
- held after storage at an internal temperature not to exceed 40°F when further processing is resumed.

3.2.2 Water. Water used for formulation, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Eggs, whole, liquid or frozen. Whole eggs may be liquid or frozen and shall have been processed in accordance with the Regulations Governing the Inspection of Eggs and Egg Products (7 CFR Part 59). The whole eggs shall be egg yolks and egg whites in their natural proportions as broken directly from the shell eggs, as evidenced by a USDA Egg Products Inspection Certificate. For liquid whole eggs, the certificate shall state the date and time of pasteurization. Liquid whole eggs shall be held at an internal temperature of 28° to 40°F for not more than 72 hours from time of pasteurization until the start of formulation of the product in which they are used. Frozen whole eggs shall be held at 10°F or lower and used within 120 days. The whole eggs shall be free from sulphide-like, acidic, metallic, musty, sour, or fruity odors or flavors or other off or foreign odors and flavors, and shall be free from foreign materials.

3.2.4 Tomato paste. Tomato paste shall be U.S. Grade A or better of the U.S. Standards for Grades of Tomato Paste and shall be of the latest season's pack. See formulation (3.3) for required soluble solids content.

3.2.5 Bread crumbs. Bread crumbs shall be made from fresh white bread only. The bread crumbs shall be coarse ground, free flowing, and free from burnt, rancid, moldy, or musty odors or flavors.

3.2.6 Protein, soy, powdered. The powdered soy protein shall be derived from either soy protein concentrate or soy protein isolate and shall have the following characteristics:

Chemical characteristics

a.	Protein (N x 6.25) m.f.b.	69.0% (min)
b.	Moisture	7.0% (max)
c.	Fat (AOCS Method EE3-49)	2.0% (max)
d.	Ash	7.0% (max)
e.	pH	6.8 to 7.1
f.	Nitrogen solubility index	50.0 (min)

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Physical requirements

Particle size. At least 99.0 percent shall pass through a U.S. Standard No. 100 sieve.

Sensory characteristics

a. Shall receive a flavor score of 6.0 or higher when tested in accordance with the provisions of the USDA Northern Regional Research Center Odor and Flavor Intensity Scoring Method of the Cereal Panel Odor and Flavor Evaluation System.

b. Shall have no added flavor.

3.2.7 Onions, dehydrated, chopped. The dehydrated chopped onions shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.8 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.9 Celery, dehydrated, sliced. Dehydrated celery slices shall be obtained from clean, sound Pascal type celery. The slices shall be approximately 3/8 inch cross-cut stalk slices with no leaf cuts. The dehydrated slices shall be free-flowing, shall possess a uniform bright color, and shall have a moisture content of not more than 5.5 percent.

3.2.10 Salt. Salt shall be noniodized, white, refined, sodium chloride with or without anticaking agents and shall comply with the purity standards for sodium chloride of the Food Chemicals Codex.

3.2.11 Sugar, white, granulated. Sugar shall be white, refined, sucrose, granulated cane or beet or a combination thereof and shall comply with MIL-STD-900.

3.2.12 Monosodium glutamate (MSG). Monosodium glutamate shall comply with the Food Chemicals Codex.

3.2.13 Garlic powder. Garlic powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.14 Pepper, black, ground. The ground black pepper shall have been ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic, penetrating odor, a hot biting pungent flavor, and a light gray to speckled black-gray color. The black pepper shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that 95 percent shall pass through a U.S. Standard No. 16 sieve.

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3.2.15 Celery seed, ground. Ground celery seed shall be prepared from the seed of Apium graveolens L. The ground celery seed shall be light to rich brown in color and possess a characteristic celery odor and flavor with a warm to slightly bitter after taste. The celery seed shall contain not less than 2.0 mL of volatile oil per 100 grams and be of such size that not less than 95 percent, by weight, shall pass through a U.S. Standard No. 55 sieve.

3.2.16 Allspice, ground. Ground allspice shall be derived from dried, nearly ripe fruit of Pimenta officinalis Lindl and shall have a fragrant, clove-like aroma, strongly aromatic, pungent, clove-like flavor and a dark reddish-brown color. Allspice shall contain not less than 3.0 mL of volatile oil per 100 grams of ground allspice with 80.0 percent eugenol as a principal constituent. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 25 sieve.

3.2.17 Bay leaves, ground. Ground bay leaves shall be derived from the dried leaves of Laurus nobilis L. and shall possess a pleasant, aromatic odor and pungent, mildly bitter flavor with a pale green to yellow-green color. The volatile oil content shall be not less than 1.0 mL per 100 grams of ground bay leaves. A minimum of 95 percent, by weight, shall pass through a U.S. Standard No. 30 sieve.

3.2.18 Mushrooms, canned, stems and pieces. Canned mushroom stems and pieces shall be U.S. Grade B or better in accordance with the U.S. Standards for Grades of Canned Mushrooms.

3.2.19 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified maize food starch for use in thermostabilized foods and shall comply with MIL-STD-900. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked starch shall be bland with essentially no cereal or starch taste.

3.2.20 Oil, vegetable. The vegetable oil (excluding coconut oil and palm kernel oil) shall have a stability of at least 15 hours (Active Oxygen Method), a free fatty acid content not to exceed 0.05 percent by weight, and a moisture/volatile matter content not to exceed 0.06 percent by weight and shall pass the cold test of 5.5 hours (minimum).

3.2.21 Caramel color. Caramel color shall comply with the Food Chemicals Codex.

3.2.22 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

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3.2.23 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements of this document. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef meeting the requirements of 3.2.1.1 shall be further trimmed, if necessary, so as to ensure compliance with finished product requirements (see 3.6). The chilled boneless beef shall be ground once through a plate with holes measuring 1 inch in diameter. The ground beef shall be held for not more than 24 hours at an internal temperature of 28° to 40°F prior to meatloaf preparation.

3.3.2 Preparation of the meatloaf mixture. The meatloaf shall be formulated and prepared as follows:

Ingredients	Percent by weight
Beef, ground	70.94
Water	11.45
Eggs, whole	4.00
Tomato paste (24% natural tomato soluble solids) 1/	3.00
Bread crumbs	2.00
Protein, soy, powdered	2.00
Onion, dehydrated, chopped	2.00
Onion powder	1.50
Celery, dehydrated, sliced	1.00
Salt	0.85
Sugar, granulated	0.50
MSG	0.50
Garlic powder	0.08
Pepper, black, ground	0.07
Celery seed, ground	0.05
Allspice, ground	0.03
Bay leaves, ground	0.03

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solid level to that specified.

a. All the ingredients shall be combined and mixed only to the extent necessary to produce a uniform mixture of ingredients.

b. The uniform mixture of ingredients shall be ground once through a plate having holes 1/4 inch in diameter without further mixing prior to stuffing (see 3.3.3).

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3.3.3 Stuffing and forming. The meatloaf mixture shall be immediately mechanically stuffed into number 7 prestuck E-Z-peel casings. The stuffed meatloaf logs shall be formed, by pressing between spring loaded screens separated by 1-7/8 inch spacers, into rectangularly shaped loaves that are 1-5/8 to 1-7/8 inches in height.

3.3.4 Cooking and cooling. The stuffed and formed meatloaf shall be maintained in the temperature range of 28° to 40°F for not more than 4 hours from time of stuffing to start of cooking. The loaves shall be dry heat cooked at 190° to 200°F until heated to an internal temperature of 160° to 170°F. The cooked loaves shall be cooled immediately to an internal temperature of 28° to 40°F and maintained at this temperature for not more than 48 hours prior to tray pack filling.

3.3.5 Slicing. The cooked cooled meatloaves shall be uniformly sliced into slices that will meet the requirements in 3.4 and 3.6.

3.3.6 Preparation of the gravy. The gravy shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	72.83
Tomato paste (24% solids) 1/	11.00
Mushrooms, canned	6.00
Starch, modified, high opacity	5.50
Oil, vegetable	1.00
Onion powder	1.00
Salt 2/	0.85
Sugar, granulated	0.50
MSG	0.50
Color, caramel	0.35
Lecithin	0.25
Pepper, black, ground	0.07
Garlic powder	0.06
Allspice, ground	0.03
Bay leaves, ground	0.03
Celery seed, ground	0.03

1/ If the specified tomato solids are not available, the tomato solids that are used must be calculated to bring the solids level to that specified.

2/ The total amount of salt in the gravy formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6r).

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NOTE: The following gravy preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified for the prepared gravy are still applicable.)

a. The mushrooms and brine from the cans shall be blended together to produce a smooth, homogenized mixture.

b. A slurry shall be made by mixing the starch and part of the water.

c. The mushroom mixture and the remaining ingredients shall be mixed together and heated to approximately 165° to 170°F. The slurry shall be added and the final mixture, heated to 165° to 170°F, and held for approximately 5 minutes at this temperature.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The gravy shall be maintained in the temperature range of 150° to 170°F for not more than 4 hours prior to tray pack filling.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with the product such as to conform to the finished product requirements and to the following requirements:

a. Not less than 36 slices of meatloaf shall be placed (shingled) in two rows lengthwise in the can (see 5.1.1) and, the can immediately filled with gravy to not less than 104 ounces of product.

b. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirements (see 3.6u).

c. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

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d. There shall be not less than 36 uniform intact slices per tray pack can.

e. The average drained weight shall be not less than 54.0 ounces.

f. The drained weight of 36 slices in an individual can shall be not less than 52.0 ounces.

g. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in a can shall be not greater than 1 ounce.

h. No individual can shall contain a bone piece measuring 0.3 inch or more in any dimension.

i. No individual can shall have a meat slice with adherent fat or a fat pocket measuring 1/2 inch or more in any dimension.

j. Texture of the meatloaf shall not be dry, tough, rubbery, or mushy.

k. The gravy shall not be lumpy.

l. Gravy shall be light brown to dark brown in color.

m. Slices shall be uniform in shape and size.

n. The average net weight shall be not less than 104 ounces.

o. No individual can shall have a net weight of less than 102 ounces of product.

p. The average fat content of the finished product shall be not greater than 8.0 percent.

q. No individual tray pack shall have a fat content greater than 10.0 percent.

r. The salt content of the finished product shall be not greater than 1.3 percent or less than 0.5 percent in any individual can.

s. The product shall show no evidence of excessive heating (materially darkened or scorched).

t. The viscosity of the gravy shall be not less than 4.5 cm per 10 seconds nor greater than 12.0 cm per 10 seconds when determined by a Bostwick Consistometer, or not less than 18,000 centipoise nor greater than 70,000 centipoise when determined by a Brookfield Viscosimeter (see 4.5.3.4).

u. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

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3.6.1 Palatability. The finished product shall be equal to or better than the preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.5.1.1 Beef examination for condition and cut. All beef shall be examined in either the bone-in or boneless state for conformance to the condition and cut requirements in 3.2.1. Cuts initially examined in the boneless state shall be in the form of whole boneless recognizable cuts or portions thereof. Any nonconforming beef shall be rejected.

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4.5.1.2 Boned and trimmed beef examination. After boning and trimming and prior to any further processing or to any freezing, the beef shall be examined for the defects listed in table II. The lot size, expressed in terms of pounds, shall be declared to the Agricultural Marketing Service (AMS) agent by the contractor. However, the AMS agent reserves the right to declare as a lot a portion of a declared lot, if, in his or her opinion, that portion may be out of compliance with any requirement. The sample unit shall be a minimum of 12 pounds of adjacent boneless beef. The sample size shall be as specified in table I. If all or a portion of the sample unit falls within a larger cut, the entire cut shall be examined. Failure of the beef to meet the acceptance criteria as indicated in table I shall be cause for rejection of the lot. Except for beef rejected because of freezing, defrosting, or not being in excellent condition, the beef may be reworked by the contractor and reoffered for examination. For reexamination, the sampling plan used shall be the one in table I designed for the next larger lot size than the one under which the lot was initially rejected. Beef shall not be reexamined more than one time.

TABLE I. Sampling plan for boned and trimmed beef

Lot size (pounds)	Sample size (sample units)	Defect categories			
		Major		Minor	
		AC	RE	AC	RE
500 or less	20	2	3	5	6
501 to 1,200	32	3	4	7	8
1,201 to 3,200	50	5	6	10	11
3,201 to 10,000	80	7	8	14	15
10,001 to 38,000	125	10	11	21	22
Reinspection lots of 10,001 to 38,000	200	14	15	21	22

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TABLE II. Boned and trimmed beef defects 1/ 2/

Category		Defect
Major	Minor	
101		Presence of popliteal, prescapular, prefemoral or any exposed lymph gland measuring 0.5 inch or more in any dimension
102		Presence of blood clot measuring 1.0 inch or more in any dimension
	201	Presence of blood clot measuring 0.5 inch or more but less than 1.0 inch in any dimension
103		Presence of bruise measuring 1.0 inch or more in any dimension
	202	Presence of bruise measuring 0.5 inch or more but less than 1.0 inch in any dimension
104		Presence of bone measuring 0.3 inch or more in any dimension
105		Presence of cartilage measuring 0.5 inch or more in any dimension
106		Presence of backstrap measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	203	Presence of backstrap measuring 0.5 inch or more but less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
107		Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or the gracilis membrane) measuring 2.0 square inches or more
	204	Presence of heavy connective tissue (for example, on the surface of the outside round adjacent to the knuckle, along the skin surface of the strip loin, or gracilis membrane) measuring 0.5 square inch or more but less than 2.0 square inches

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TABLE II. Boned and trimmed beef defects 1/ 2/ - Continued

Category		Defect
Major	Minor	
	205	Presence of heavy connective tissue on lower edge of short plate or flank
	206	Presence of heavy connective (abdominal tunic) tissue on the flank measuring 1.0 inch or more in any dimension
	207	Presence of membranous portion of diaphragm or membranous covering from skirt, flank, or abdominal section of short plate measuring 3.0 square inches or more
108		Presence of knuckle cover
109		Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring 1.0 inch or more in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	208	Presence of kidney, pizzle eye, prepubic tendon, thymus gland, or hanging tender measuring less than 1.0 inch in one dimension and 0.2 inch or more in a second dimension (when measured at right angles to each other)
	209	Presence of calcified (scratchy) periosteum measuring 2.0 square inches or more
	210	Presence of shank, clod, knuckle, or bottom (outside) round with tendinous end showing less than 75 percent lean tissue on a cross-sectional cut surface
	211	Presence of dehydrated surface measuring 1.0 square inch or more
	212	Presence of discolored meat (including blood discolored neck meat) measuring 1.0 square inch or more
	213	Presence of exposed blood vessel measuring 1.0 inch or more in any dimension
	214	Presence of cod, udder, kidney, or pelvic fat

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1/ Determination of wholesomeness and acceptability of product with respect to the presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal) shall be made by a Meat and Poultry Inspection Operations employee.

2/ Evidence of freezing or defrosting or product not in excellent condition shall cause rejection of the lot.

4.5.1.3 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 104 ounces or if the net weight of any individual can is less than 102 ounces, the lot shall be rejected.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table III. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated for 35 minutes in boiling water, opened, and inspected for the defects listed in table IV.

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TABLE III. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3,200	8	-	0	2
	8	16	1	2
3,201 to 35,000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and, if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Product defects 1/ 2/

Category	Defect
Major	
101	Less than 36 intact slices in a can
102	Drained weight of 36 meatloaf slices in an individual can is less than 52.0 ounces 3/ 4/
103	Total weight of cartilage, coarse connective tissue, section of tendon or ligaments, and glandular material, collectively, in a can is greater than 1 ounce

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TABLE IV. Product defects 1/ 2/ - Continued

Category	Defect
Major	
104	Presence of bone measuring 0.3 inch or more in any dimension
105	Meatloaf slice in a can has adherent fat or a fat pocket measuring 1/2 inch or more in any dimension
106	Meatloaf slices dry, tough, rubbery, or mushy
107	Slices not uniform in shape or size
108	Gravy is lumpy
109	Gravy not light brown to dark brown in color
110	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of foreign material (e.g., glass, dirt, insect parts, hair, wood, metal), foreign odor or flavor (e.g., burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off, strained through a U.S. Standard No. 8 sieve, and reserved for viscosity determinations. The remaining contents of the can shall be poured into a flat bottom container. A minimum of three times the tray pack can's volume of 190° to 212°F water shall be added to the container so as to cover the contents. The contents and water shall be agitated so as to liquify rendered fat and remove the gravy without breaking the meatloaf slices. The contents shall then be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the meatloaf slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate 45 degree angle and

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allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ The lot shall be rejected if the sample average drained weight is less than 54.0 ounces.

4.5.3.3 Fat and salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis.
- Three for submission to the contractor.
- Three for retention by the AMS agent as reserve samples.

The three tray packs for laboratory analyses shall be individually tested for fat and salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, chapter: Meat and Meat Products, except that preparation of the samples shall be as follows. The three unopened cans shall be gently warmed in a water bath to melt fat adhering to the inside of the cans. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. The test results shall be reported to the nearest 0.1 percent. Any test failure is a major defect. The lot shall be rejected if any of the following major defects occur:

a. The average fat content of the three tray packs is greater than 8.0 percent.

b. The fat content of any individual tray pack is greater than 10.0 percent.

c. The salt content of any individual tray pack is greater than 1.3 percent or less than 0.5 percent.

Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the fat or salt content requirement. When the reserve samples are analyzed, the analyses for both fat and salt shall be made and will be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

4.5.3.4 Viscosity testing. The strained free liquid collected from each of the cans in the first sample of cans selected for drained weight inspection (see 4.5.3.2 and 3/ to table IV) shall be individually tested for viscosity as specified in 4.5.3.4.1 or, alternatively, as specified in 4.5.3.4.2 (see 6.4).

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4.5.3.4.1 Bostwick Consistometer method.

Instrument: Bostwick Consistometer
Catalog Number: 23270-004
VWR Scientific Company
P.O. Box 7900
San Francisco, CA 94120

or

Catalog Number: 15-347-50
Fisher Scientific
585 Alpha Drive
Pittsburgh, PA 15238

Method:

a. Level the instrument.

b. Bring gravy to 100° ± 1°F in a water bath in a covered container.

c. Stir gravy thoroughly before filling the Bostwick cavity.

d. Scrape gravy evenly across upper edge of cavity.

e. Release gravy and time gravy flow to the nearest 1 second and measure distance traveled to the nearest 0.1 cm.

If the Bostwick viscosity value of the gravy from any can in the sample is less than 4.5 cm per 10 seconds or greater than 12.0 cm per 10 seconds, it is a major defect and the lot shall be rejected.

4.5.3.4.2 Brookfield Viscosimeter method.

Instrument: Brookfield Viscosimeter (variable speed) equipped with a No. 4 spindle.

Brookfield Engineering Lab, Inc.
Stoughton, MA 02072

Method:

a. Bring gravy to 80° ± 1°F in a water bath in a covered container.

b. Stir gravy thoroughly and pour 500 mL of the gravy into a 600 mL beaker.

c. Using a number 4 spindle, stir at a speed of 5.0 rpm and record dial reading.

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d. Convert dial reading to centipoise. Record result to nearest 100 centipoise.

If the Brookfield viscosity value of the gravy from any can in the sample is less than 18,000 centipoise or greater than 70,000 centipoise, it is a major defect and the lot shall be rejected.

4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers except that inspection for labeling shall be in accordance with 4.5.4.1. In addition, the following defect shall be classified as a critical defect and shall be cause for rejection of the lot:

Evidence of buffing causing damage (i.e. scratches or scuffing) to exterior can coating.

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels the following additional defects shall also apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can i.e., label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and 21 CFR, Part 113, Subpart D, or 9 CFR, Part 318, Subpart G, as applicable. Any nonconformance based on observation of can seam teardowns or on record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

4.5.6 Vacuum examination. Cans shall be allowed to cool to 75° ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, lay the straight edge only between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for the rejection of the lot.

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4.5.7 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A.

5.1.1 Level A. One hundred and four ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. Buffing of the can causing damage to the exterior can coating is not permitted (see 4.5.4).

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses

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having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (e.g., est 38) or a three-letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987, would be coded as 7078). The Julian code shall represent the day the product was

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packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of the product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT: Submerge unopened can in boiling water. Simmer gently 40 - 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place. Place in a 350°F oven 35 - 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2 slices each.

A pre-printed, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering label is used, the tray pack can shall be labeled with a Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT
OR MORE THAN TWO HIGH IN STORAGE,
UNLESS PALLET RACKS ARE USED.

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6. NOTES

6.1 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of packing required (see 5.2).
e. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article inspection. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Alternative viscosity testing. The contracting officer may authorize an alternative contractor-recommended method of viscosity testing if the alternative method is approved by the U.S. Army Natick Research, Development, and Engineering Center.

6.5 Subject term (key word) listing.

Canned foods
Combat field feeding
Meatloaf
Thermostabilized
Tray pack

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Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Preparing activity:

Army - GL

Project No. 8940-0609

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submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-44345	Meatloaf With Mushroom Gravy, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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